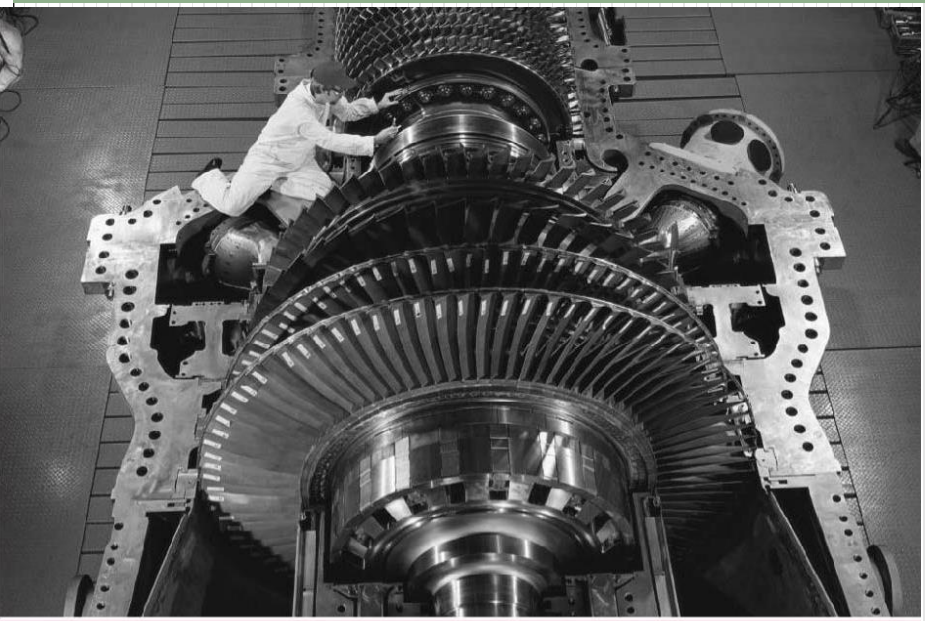
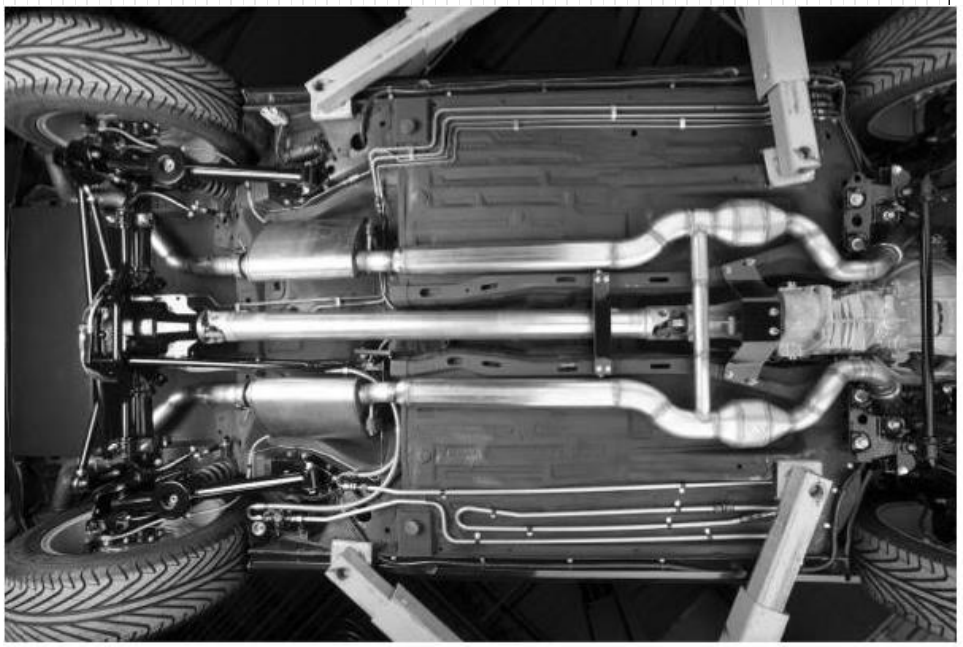
Chapter 6
TORSION

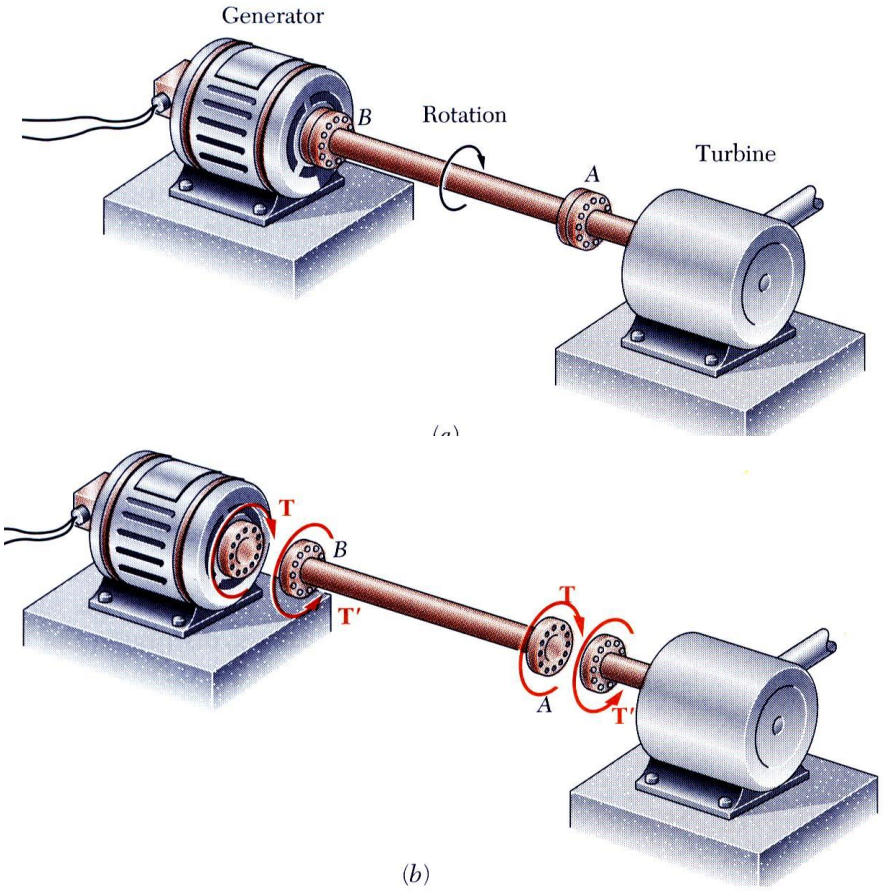


This chapter is devoted to the study of torsion and of the stresses and deformations it causes. In the jet engine shown here, the central shaft links the components of the engine to develop the thrust that propels the plane.



In the automotive power train shown, the shaft transmits power from the engine to the rear wheels.

Torsional Loads on Circular Shafts

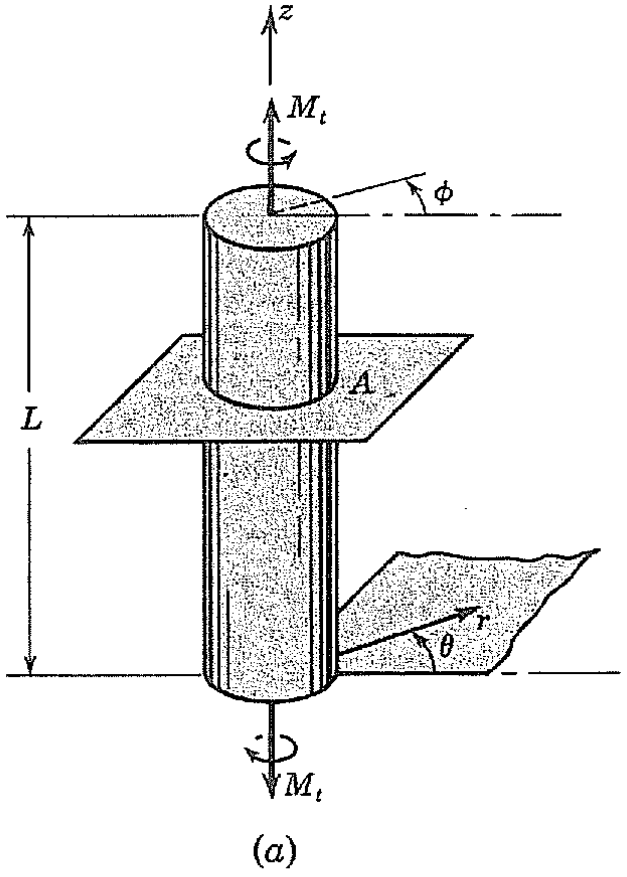


- Interested in stresses and strains of circular shafts subjected to twisting couples or *torques*
- Turbine exerts torque *T* on the shaft
- Shaft transmits the torque to the generator
- Generator creates an equal and opposite torque *T'*

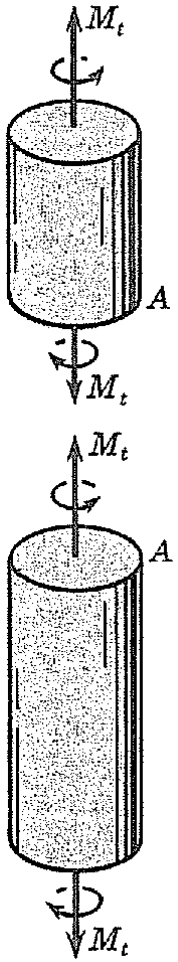
In the transmission of power by a shaft in torsion, we are interested primarily in the twisting moment which can be transmitted by the shaft without damage to the material, and hence we wish to know what the stresses are in the shaft.

Geometry of deformation of a twisted circular shaft

- Use of cylindrical coordinates r, θ, z

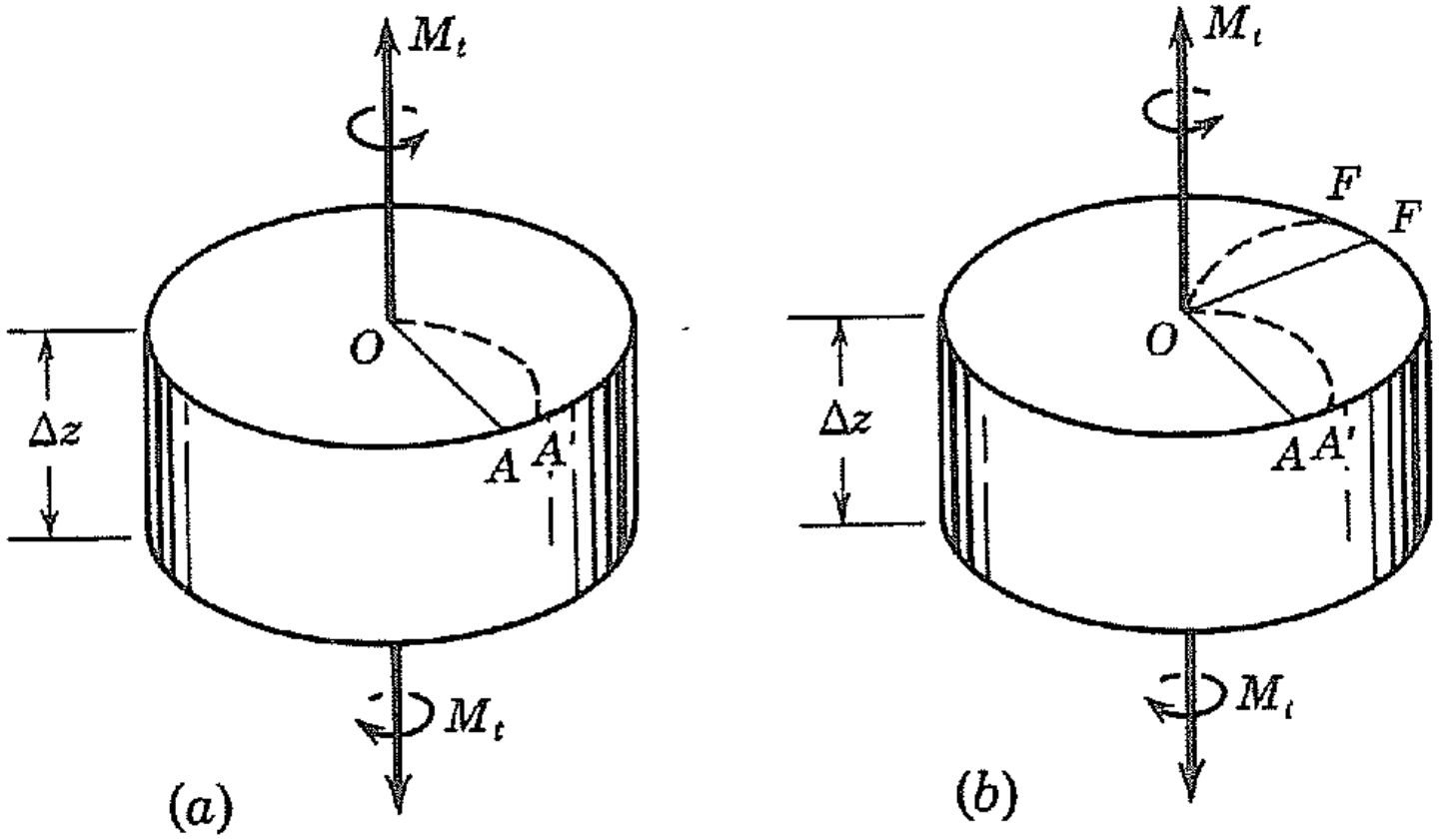


(a)

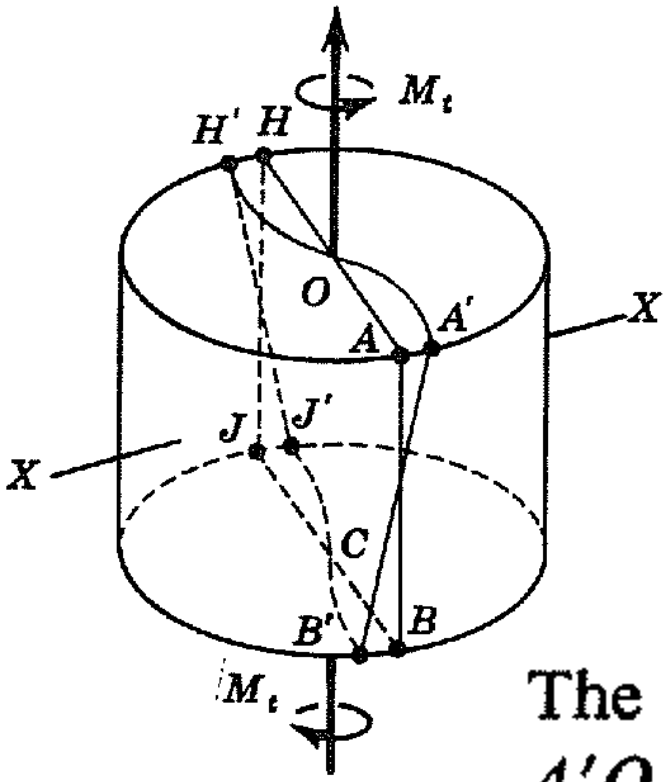


(b)

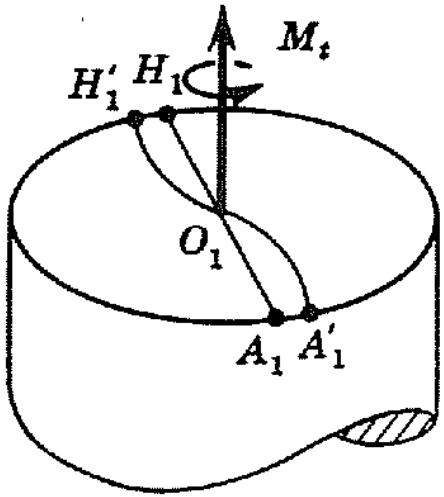
Equilibrium requires that each cross section of the shaft must transmit the twisting moment M_t.



- When a circular shaft is twisted, the cross sections must remain plane



The assumed shape of $B'CJ'$ does not match $A_1'O_1H_1'$.



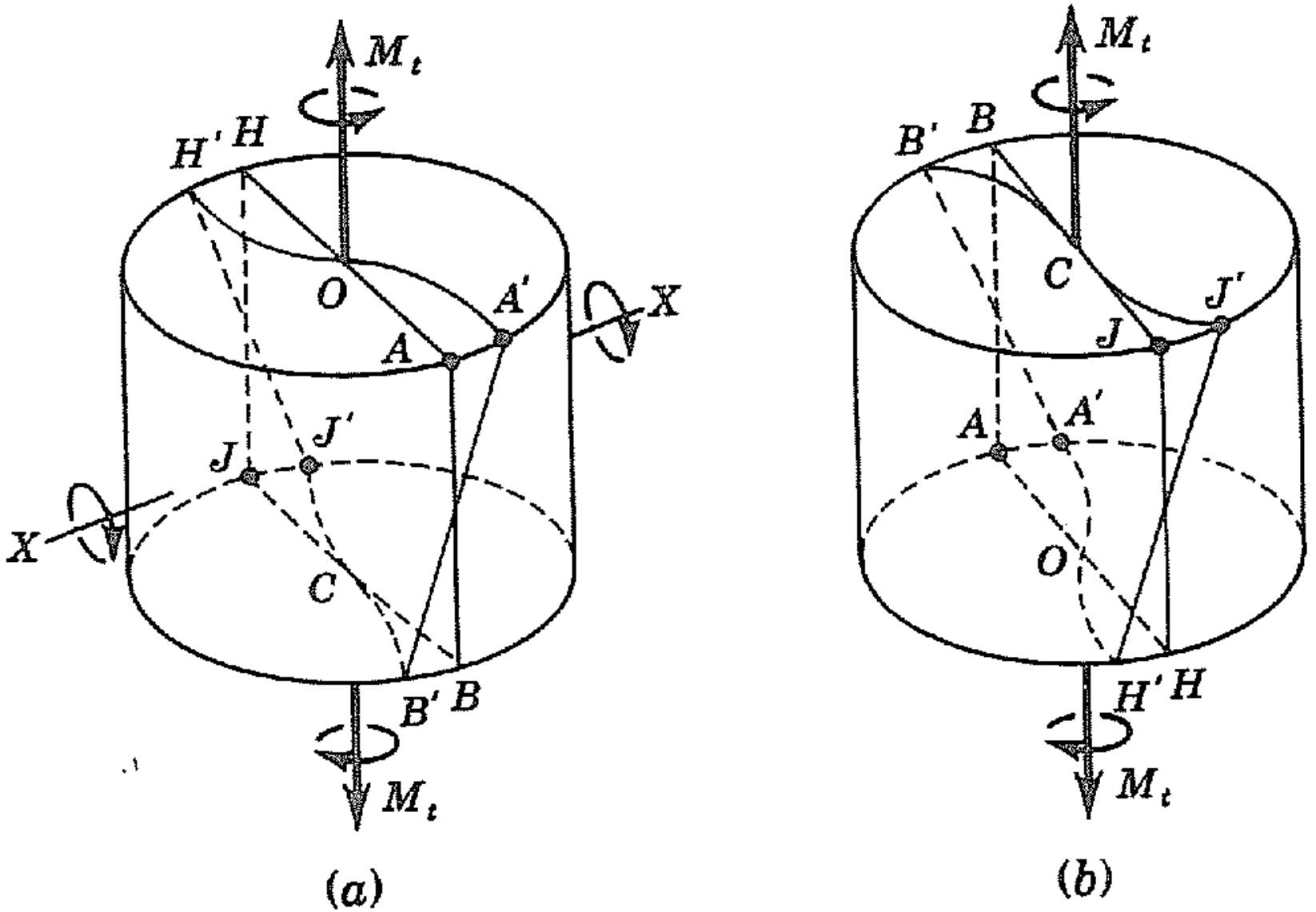


Fig. 6.6 Rotating (*a*) about *X*-*X* through 180° yields (*b*), which has undergone different deformation even though the twisting moment and geometry are the same.

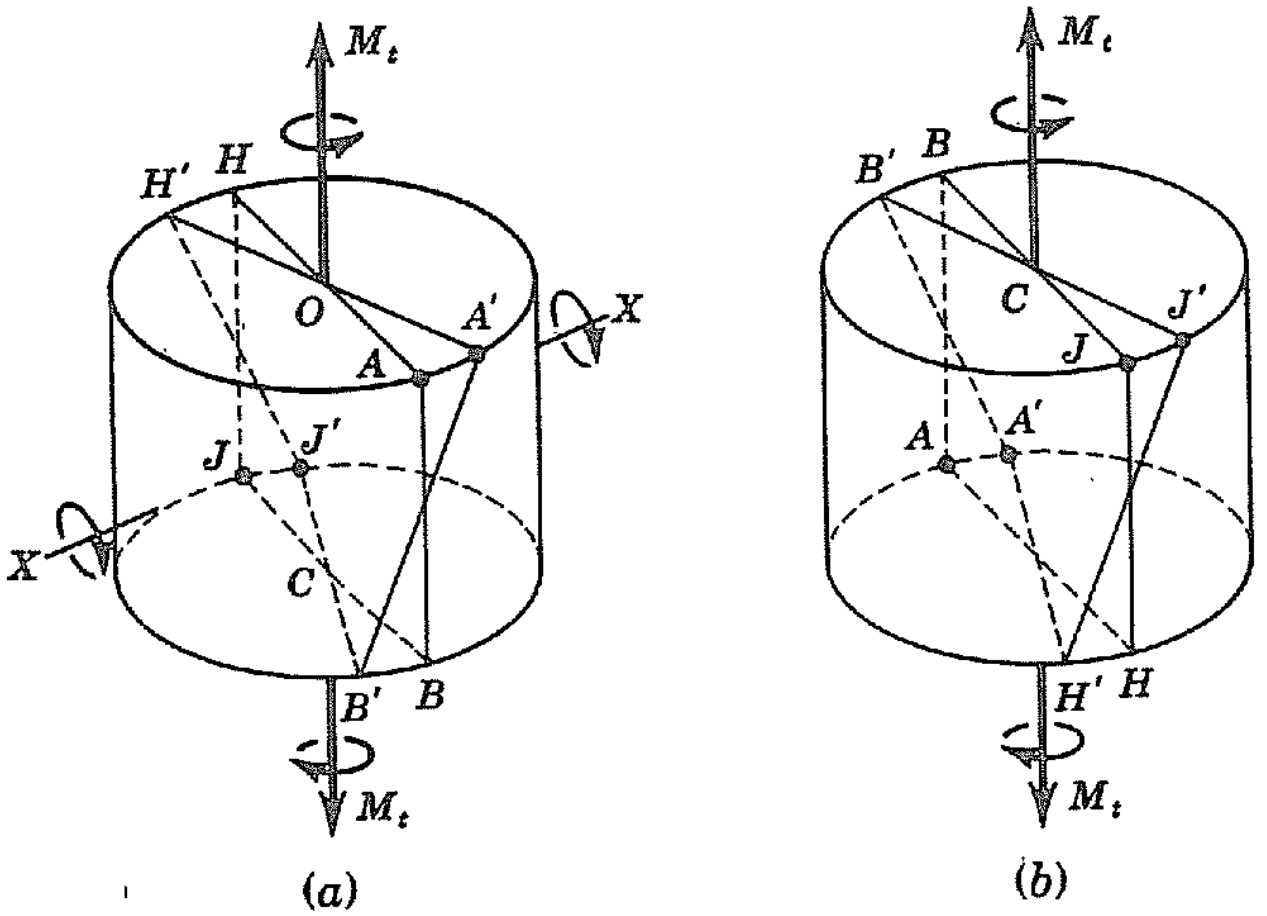


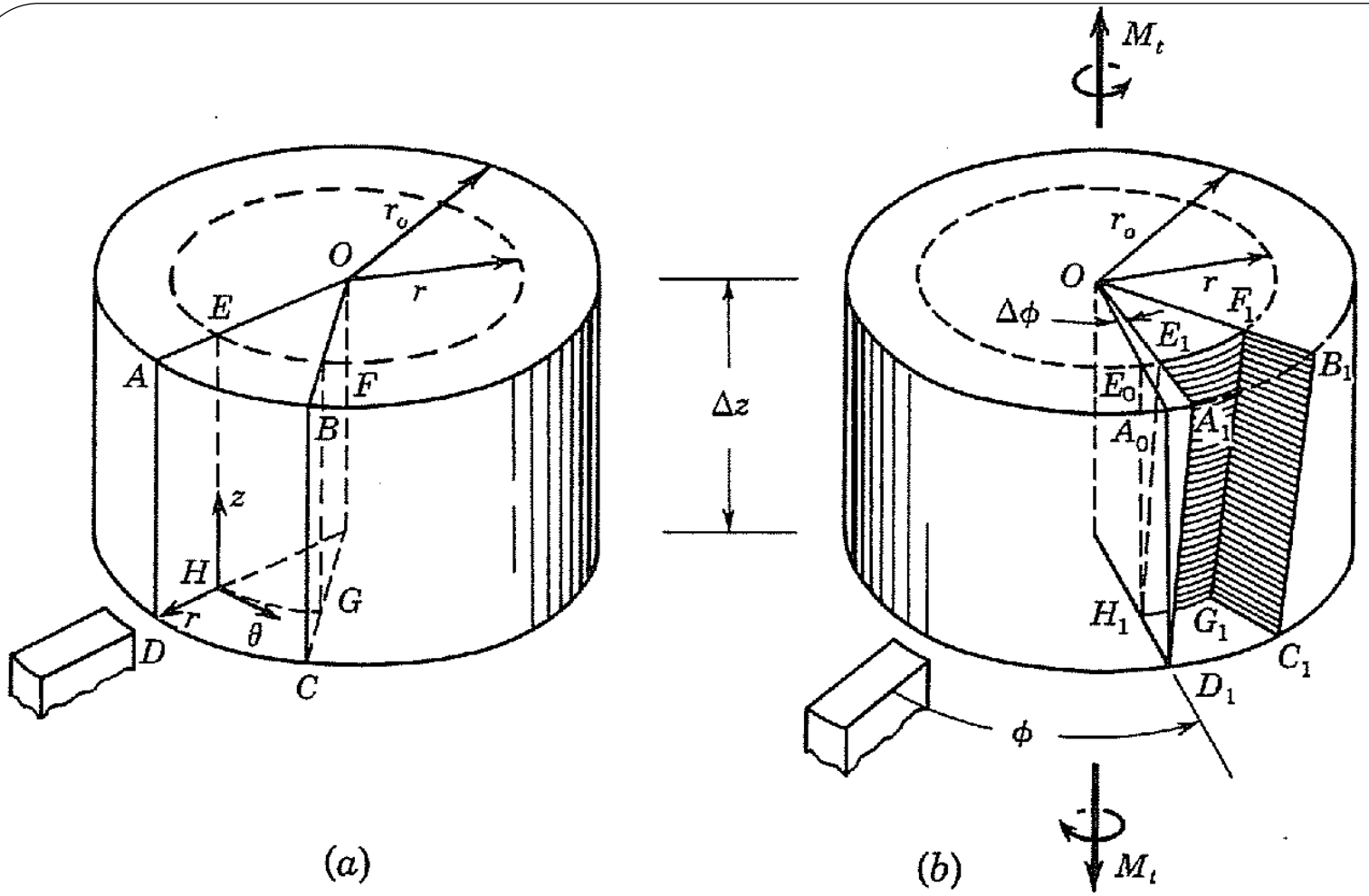
Fig. 6.7 If the diameter HA remains straight during deformation, then rotation of (*a*) about X-X produces (*b*) which is identical in terms of deformation.

- Each plane cross section originally normal to the axis remains plane and normal and does not distort within its own plane.

Dilational deformation

$$\epsilon_r = \epsilon_\theta = \epsilon_z = 0$$

With the assumption that extensional strains vanish, the only remaining possible mode of deformation is one in which the cross sections of the shaft remain undeformed but rotate relative to each other.



(a) (b)

$$\epsilon_r = \epsilon_\theta = \epsilon_z = \gamma_{r\theta} = \gamma_{rz} = 0$$

$$\gamma_{\theta z} = r\frac{d\phi}{dz}$$

$$\gamma_{\theta Z} = \lim_{\Delta Z \to 0} \frac{E_0 E_1}{H_1 E_0}$$

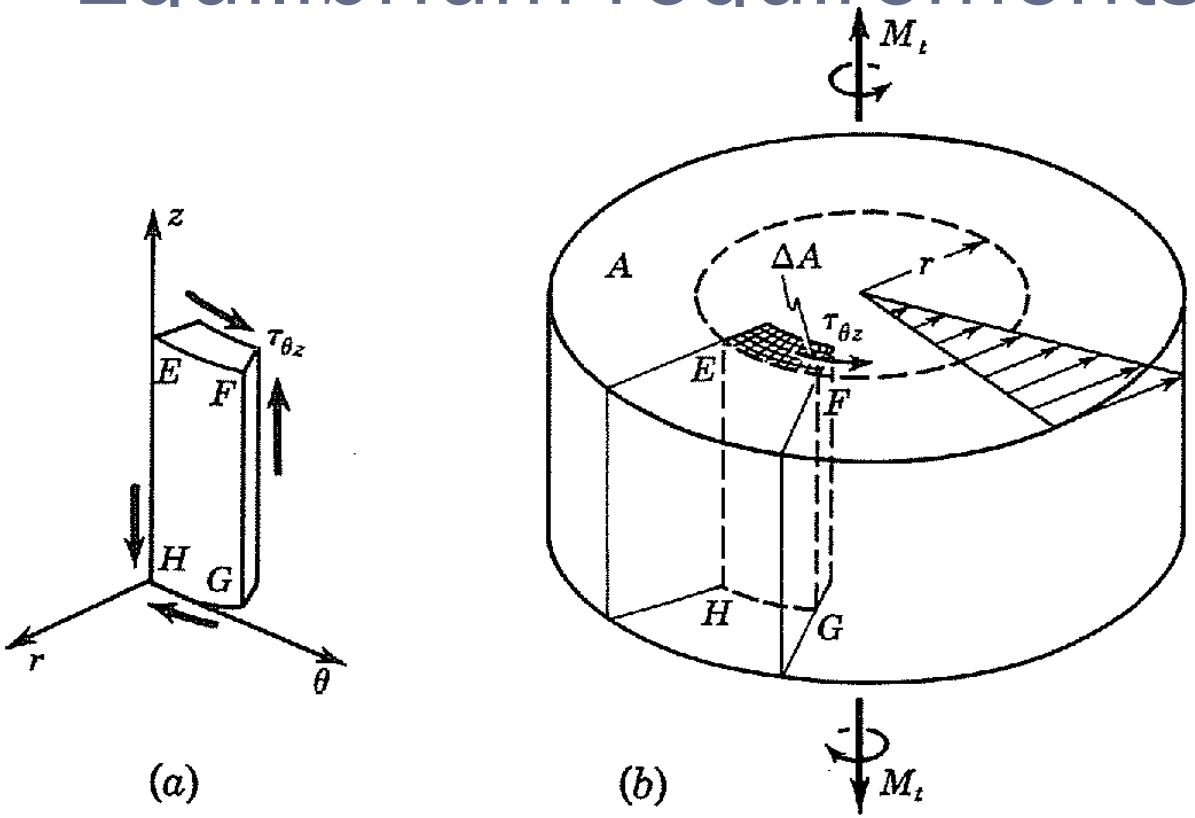
$$\gamma_{\theta Z} = \lim_{\Delta Z \to 0} \frac{r\Delta\emptyset}{\Delta Z}$$

Stresses obtained from stress strain relations

$$\sigma_r = \sigma_\theta = \sigma_z = \tau_{r\theta} = \tau_{rz} = 0$$

$$\tau_{\theta z} = G\gamma_{\theta z} = Gr\frac{d\phi}{dz}$$

Equilibrium requirements



$$\int_A r(\tau_{\theta z} dA) = M_t$$

Stress and deformation in a twisted elastic circular shaft

From

$$\tau_{\theta z} = G\gamma_{\theta z} = Gr\frac{d\phi}{dz} \quad \text{------ (A)}$$

and $$\int_A r(\tau_{\theta z}\, dA) = M_t$$

$$M_t = G\frac{d\varphi}{dz}\int_A r^2\, dA - - - - - - - - 1$$

$$M_t = G\frac{d\varphi}{dz} I_Z - - - - - - - - - - 2 \qquad I_z = \frac{\pi r^4}{2} = \frac{\pi d^4}{32} - - - - - 3$$

From Equation2

$$\frac{d\varphi}{dz} = \frac{M_t}{GI_Z} \quad ----4$$

The total angle of twist between the ends is calculated by integrating eq 4

$$\varphi = \int_0^L \frac{M_t}{GI_Z}\, dz = \frac{M_t L}{GI_Z} \quad ------5$$

Substituting eq.4 in eq. A we get

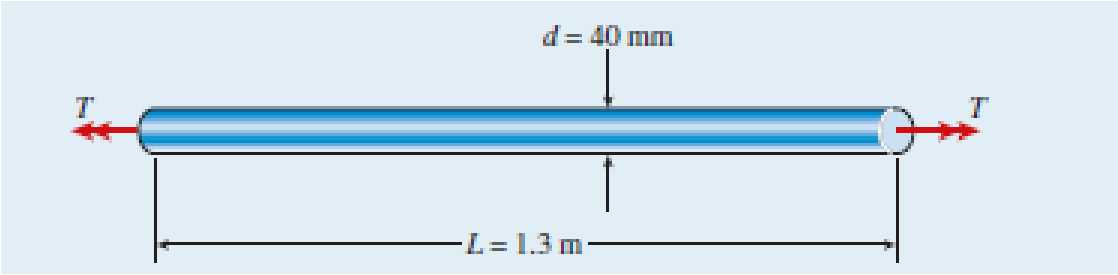
$$\tau_{\theta Z} = \frac{M_t r}{I_Z} \quad -----6$$

- $\frac{M_t}{I_P} = \frac{\tau}{r} = \frac{G\emptyset}{L}$
- $I_P = \frac{\pi d^4}{32}$ -for solid shaft
- $I_P = \frac{\pi({d_0}^4 - {d_i}^4)}{32}$ -for hollow shaft

A solid steel bar of circular cross-section has diameter d=40mm, length L=1.3m, and G=80GPa.

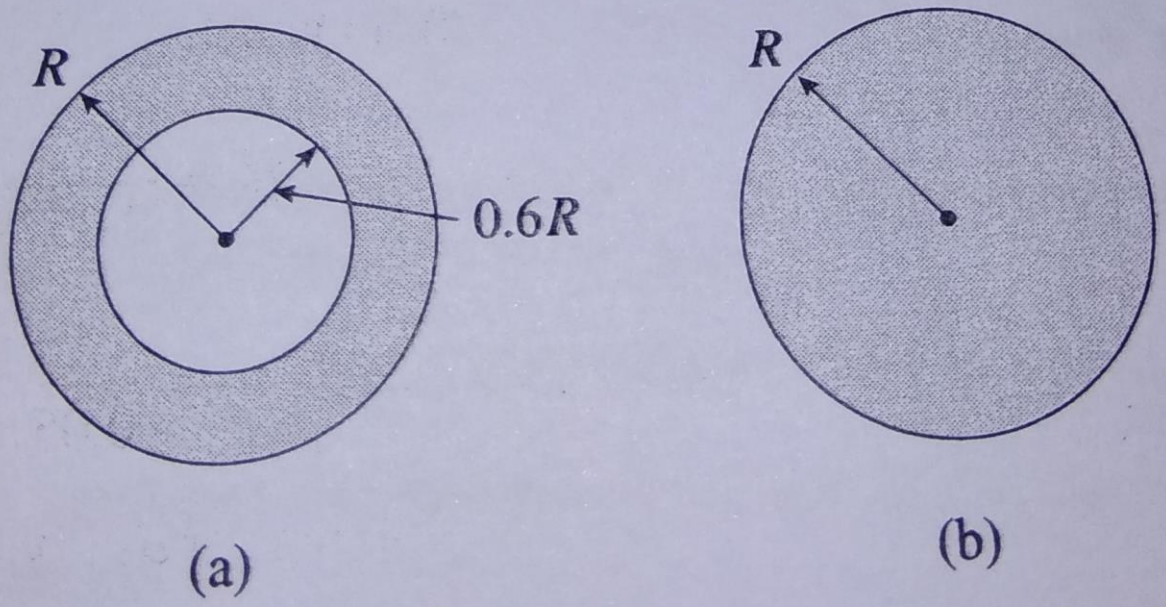
a) if T=340Nm, what is the max shear stress in the bar? What is the angle of twist between the ends.

b) if the allowable shear stress is 42MPa and the allowable angle of twist is 2.5^0, what is the maximum permissible torque?



A hollow shaft and a solid shaft constructed of the same material have the same length and the same outer radius R . The inner radius of the hollow shaft is $0.6R$.

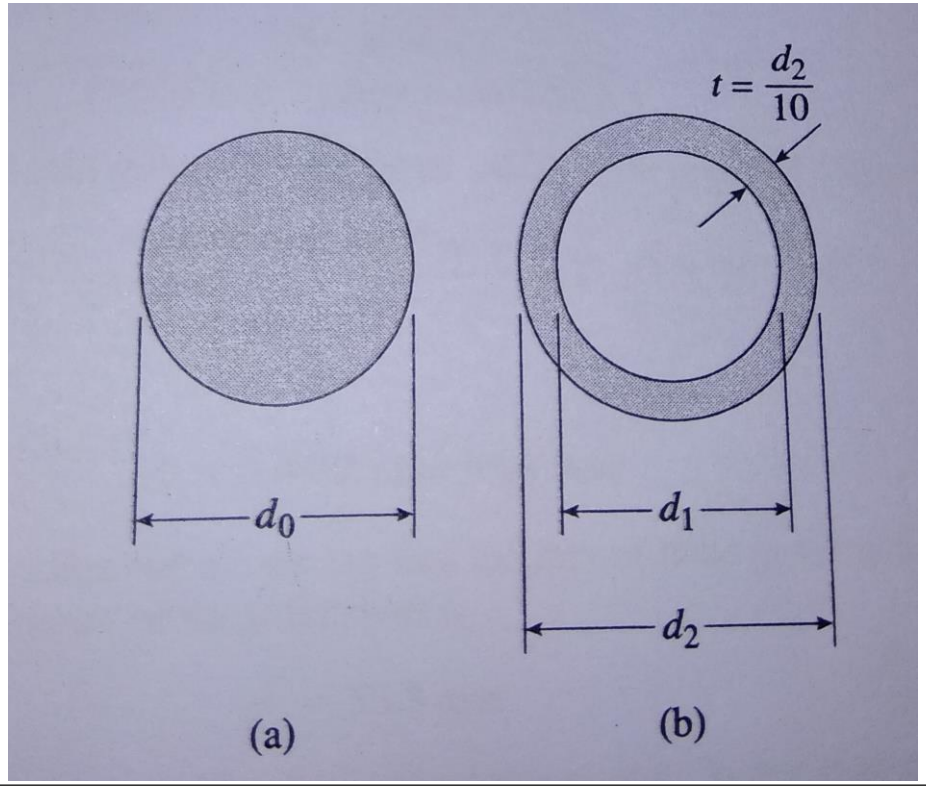
(a) Assuming that both shafts are subjected to the same torque, compare their shear stresses, angles of twist, and weights. (b) Determine the strength-to-weight ratios for both shafts.



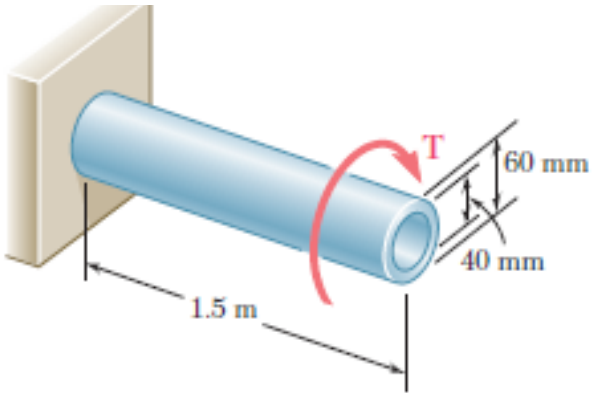
(a) (b)

A steel shaft is to be manufactured either as a solid circular bar or as a circular tube . The shaft is required to transmit a torque of 1200 N·m without exceeding an allowable shear stress of 40 MPa nor an allowable rate of twist of 0.75°/m. (The shear modulus of elasticity of the steel is 78 GPa.)

(a) Determine the required diameter d_0 of the solid shaft. (b) Determine the required outer diameter d_2 of the hollow shaft if the thickness t of the shaft is specified as one-tenth of the outer diameter. (c) Determine the ratio of diameters (that is, the ratio d_2/d_0) and the ratio of weights of the hollow and solid shafts.



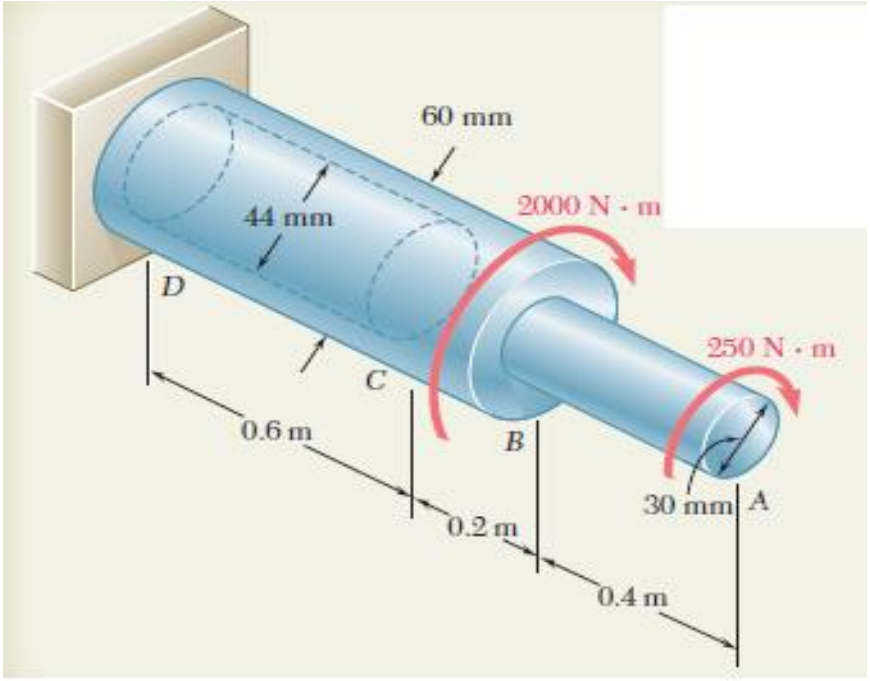
A hollow cylindrical steel shaft is 1.5 m long and has inner and outer diameters respectively equal to 40 and 60 mm . (*a*) What is the largest torque that can be applied to the shaft if the shearing stress is not to exceed 120 MPa? (*b*) What is the corresponding minimum value of the shearing stress in the shaft?

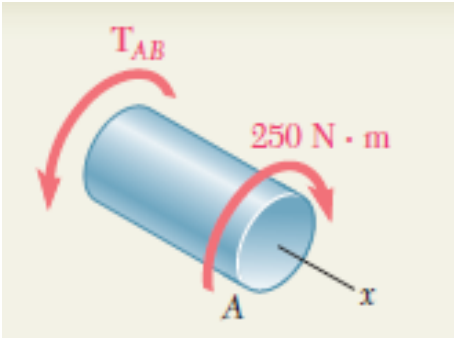


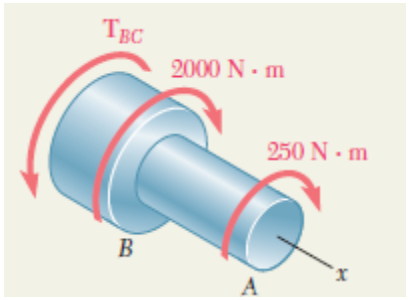
What torque should be applied to the end of the shaft of to produce a twist of 2°? Use the value $G = 77$ GPa for the modulus of rigidity of steel.

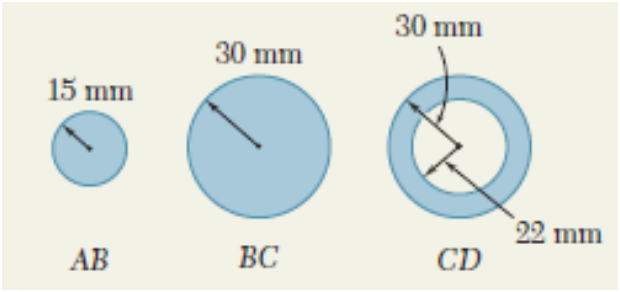
What angle of twist will create a shearing stress of 70 MPa on the inner surface of the hollow steel shaft

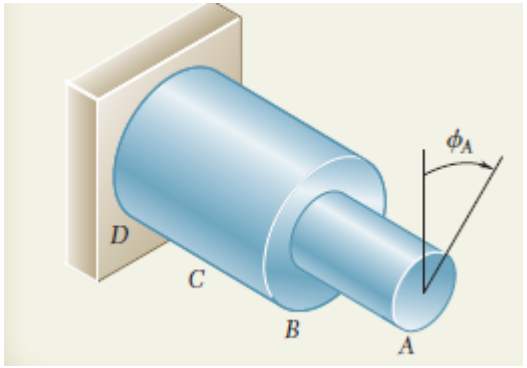
The horizontal shaft AD is attached to a fixed base at D and is subjected to the torques shown. A 44-mm-diameter hole has been drilled into portion CD of the shaft. Knowing that the entire shaft is made of steel for which $G = 77$ GPa, determine the angle of twist at end A.

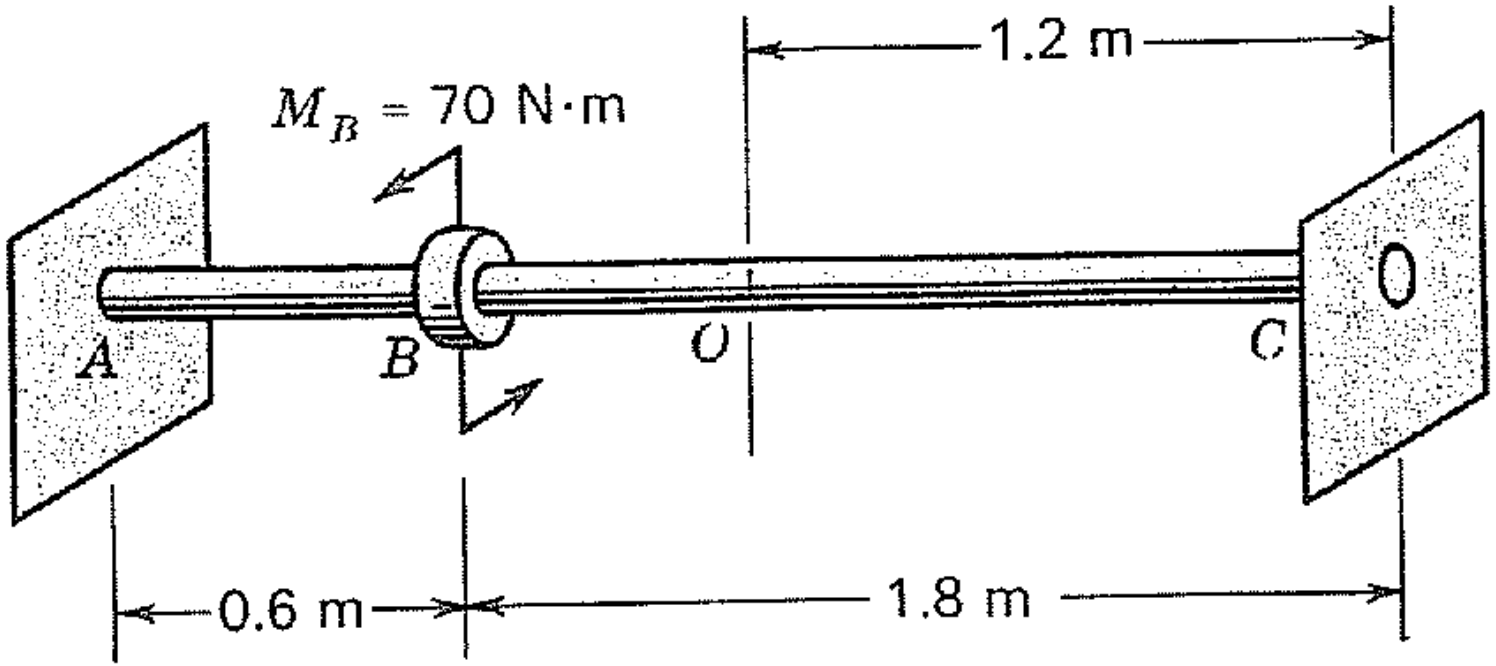



T_{AB}
250 N · m
A
x


T_{BC}
2000 N · m
250 N · m
B
A
x


15 mm
30 mm
30 mm
22 mm
AB
BC
CD


ϕ_A
D
C
B
A



- Dia = 25 mm; aluminum alloy shaft.
- Find the angle through which O rotates.
- G = 26.5 GN/m^2
- Ends A and C are built in

Stress analysis in torsion: Combined Stresses

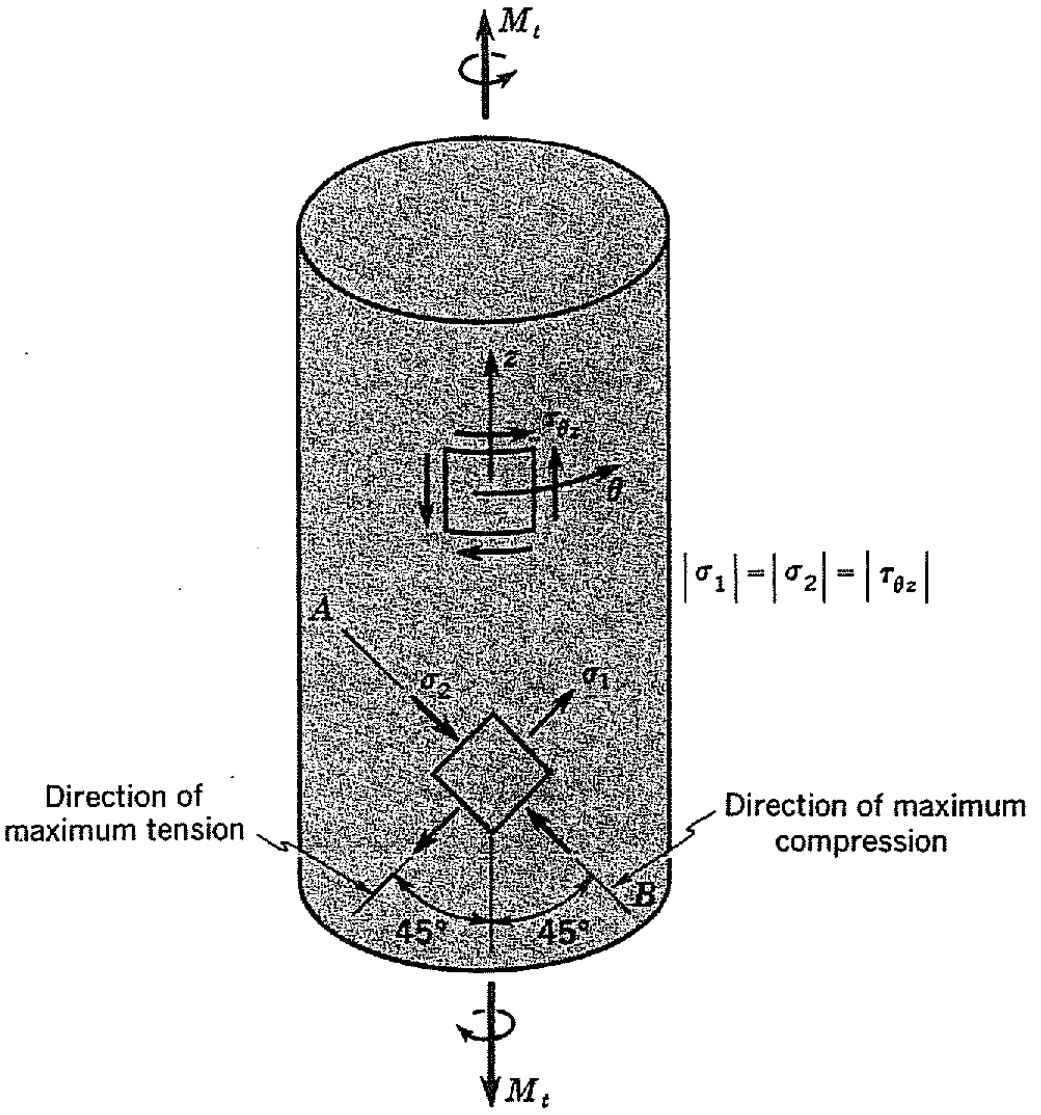
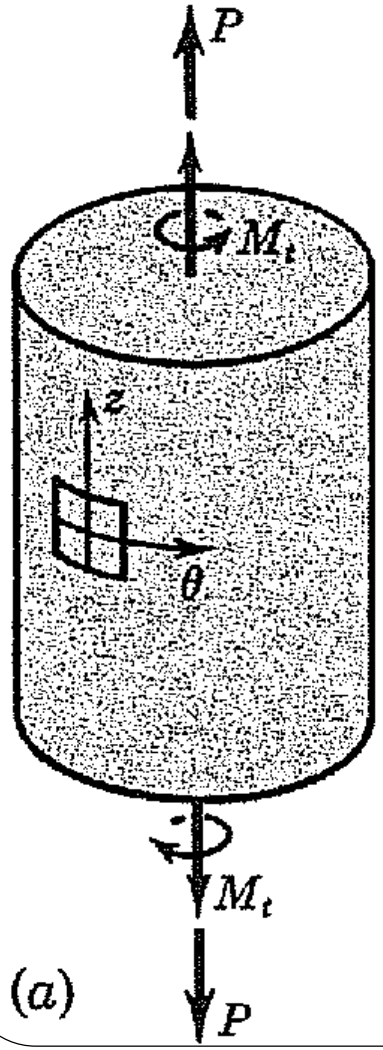
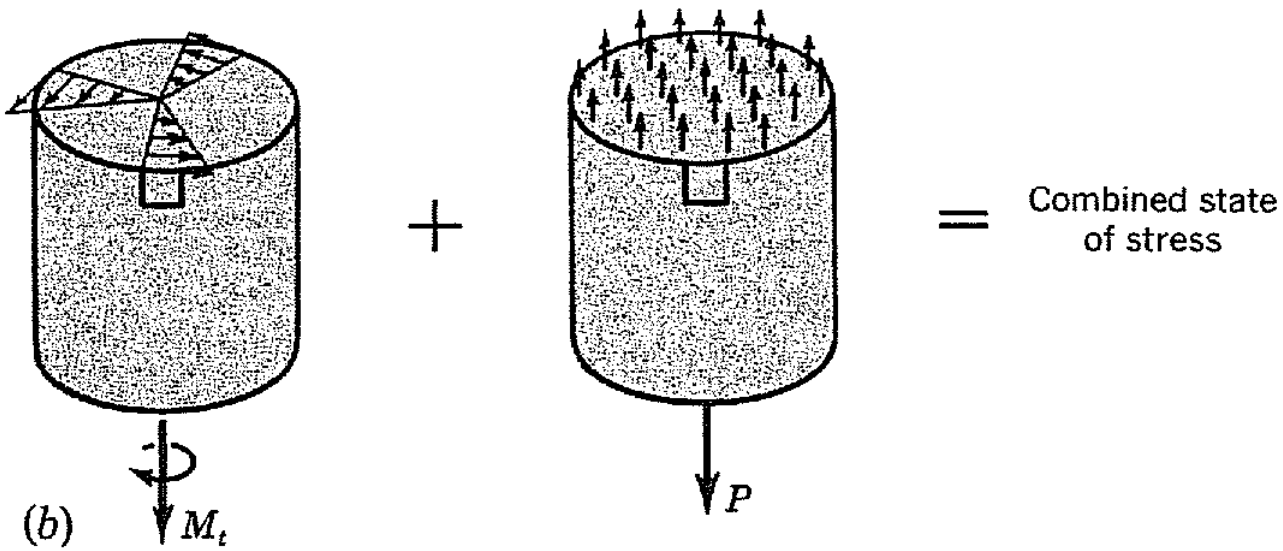


Fig. 6.14 The principal stresses in torsion are equal tension and compression acting on faces inclined at 45° to the axis of the shaft.

Stress analysis in torsion: Combined Stresses

- A uniform, homogeneous circular shaft is subjected to an axial tensile force P and a twisting moment M_t.





Strain energy due to torsion

$$U = \int_L \frac{M_t^2}{2GI_z} dz$$

The onset of yielding in torsion

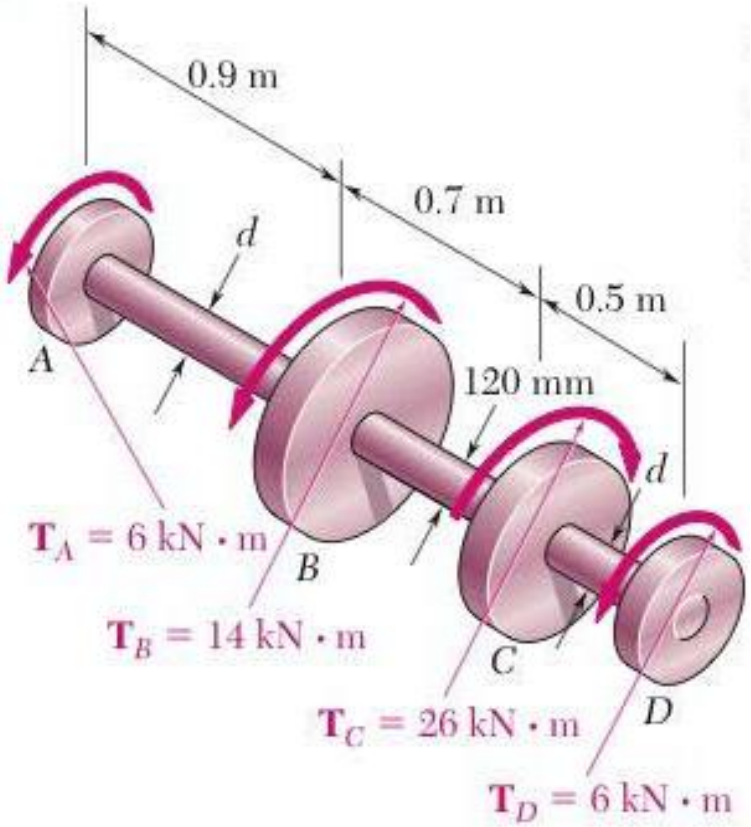
- Mises yield criterion

$$\sqrt{\tfrac{1}{2}\left[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2\right]} = Y$$

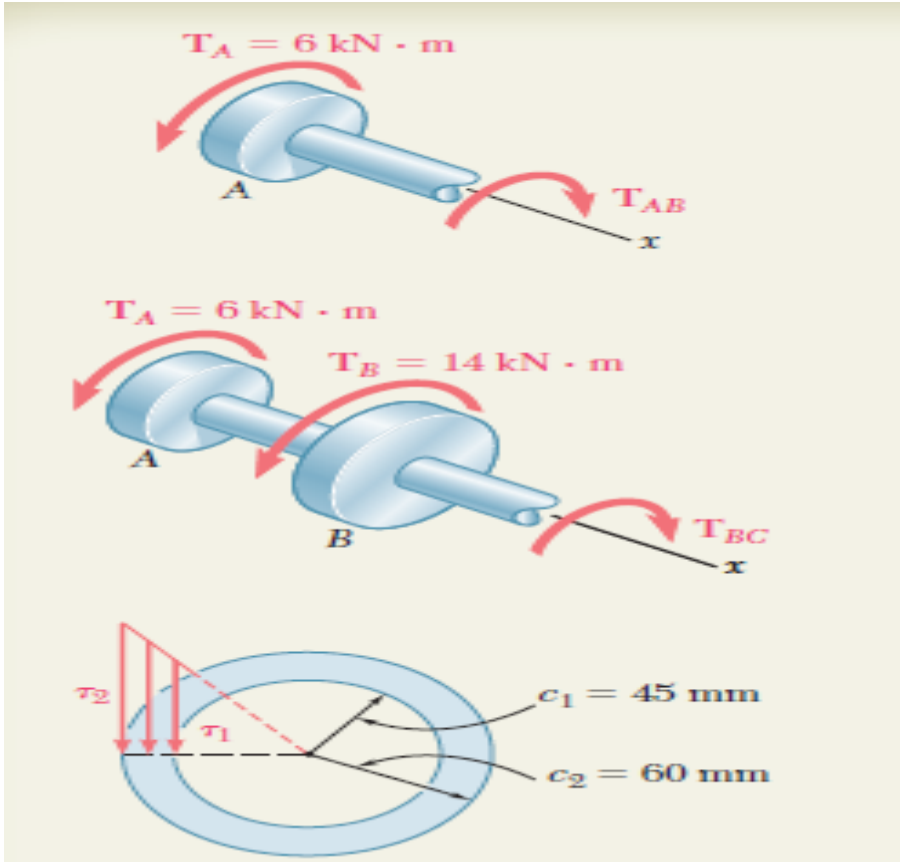
- Maximum shear-stress criterion

$$\tau_{\max} = \frac{\sigma_{\max} - \sigma_{\min}}{2} = \frac{Y}{2}$$

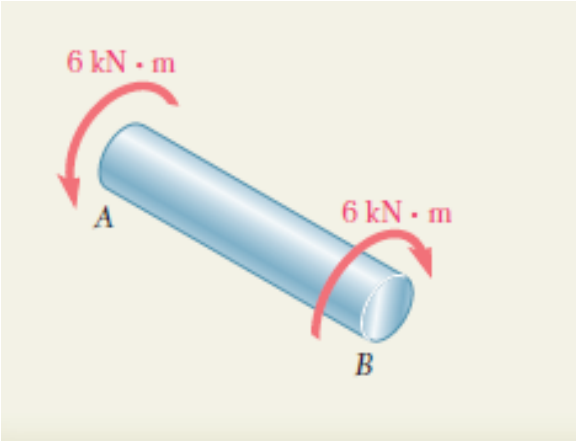
Problem



Shaft *BC* is hollow with inner and outer diameters of 90 mm and 120 mm, respectively. Shafts *AB* and *CD* are solid and of diameter *d*. For the loading shown, determine (*a*) the maximum and minimum shearing stress in shaft *BC*, (*b*) the required diameter *d* of shafts *AB* and *CD* if the allowable shearing stress in these shafts is 65 MPa.

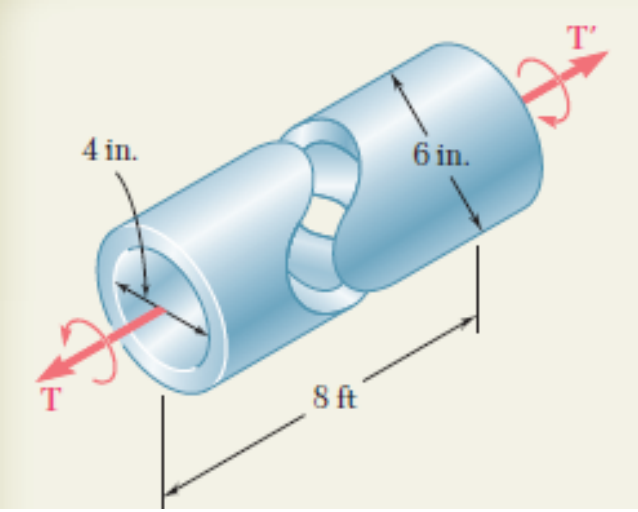

$T_A = 6$ kN · m
A
T_{AB}
x
$T_A = 6$ kN · m
$T_B = 14$ kN · m
A
B
T_{BC}
x
τ_2
τ_1
$c_1 = 45$ mm
$c_2 = 60$ mm

b

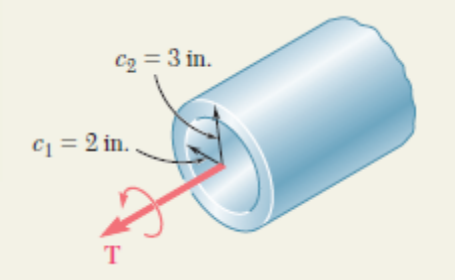


d = 77.8mm

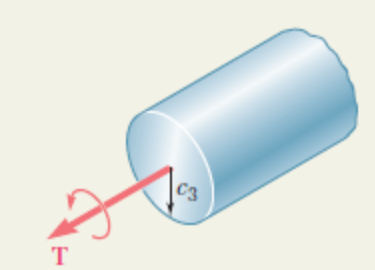
Problem



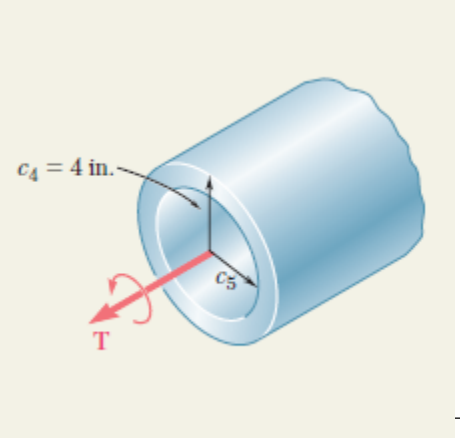
The preliminary design of a large shaft connecting a motor to a generator calls for the use of a hollow shaft with inner and outer diameters of 4 in. and 6 in., respectively. Knowing that the allowable shearing stress is 12 ksi, determine the maximum torque that can be transmitted (*a*) by the shaft as designed, (*b*) by a solid shaft of the same weight, (*c*) by a hollow shaft of the same weight and of 8-in. outer diameter.



a. Hollow Shaft as Designed.

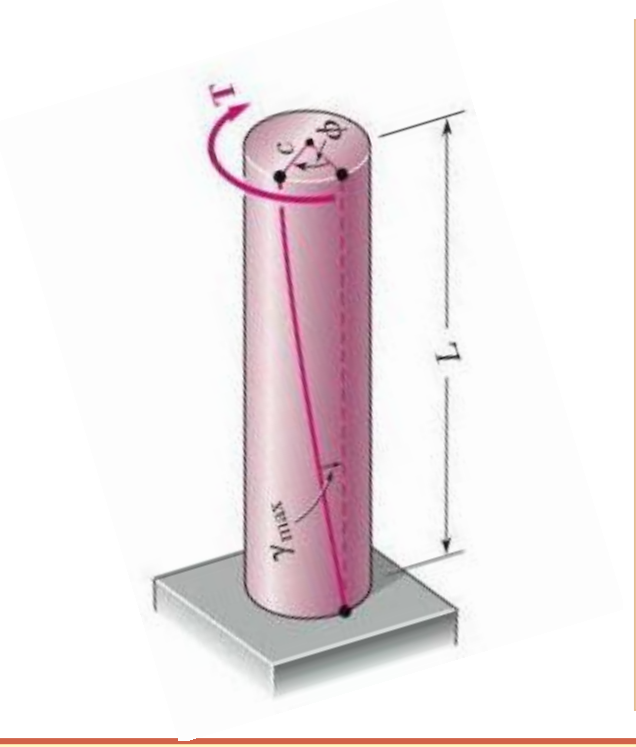


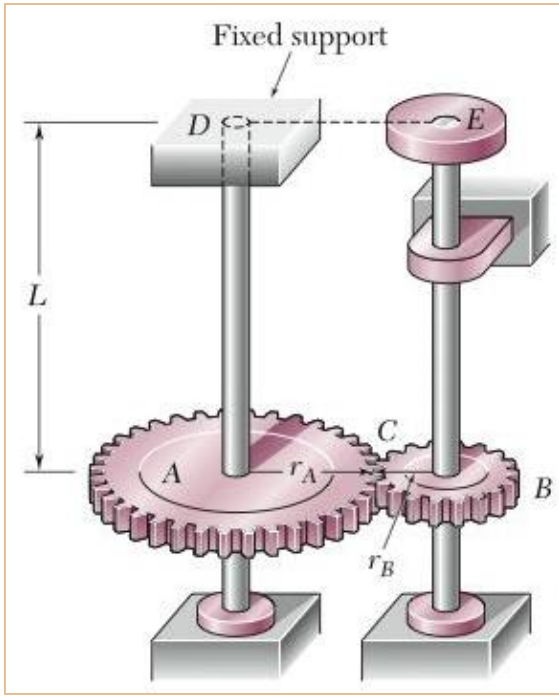
b. Solid Shaft of Equal Weight.

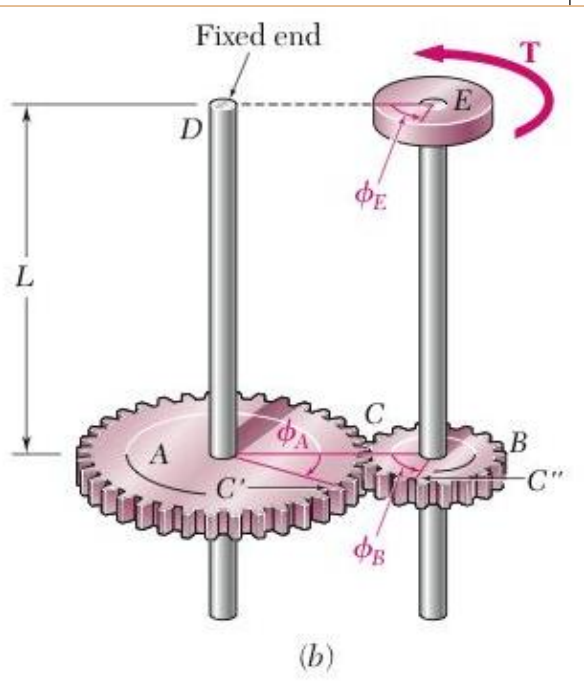


c. Hollow Shaft of 8-in. Diameter.

Angle of twist in the elastic range

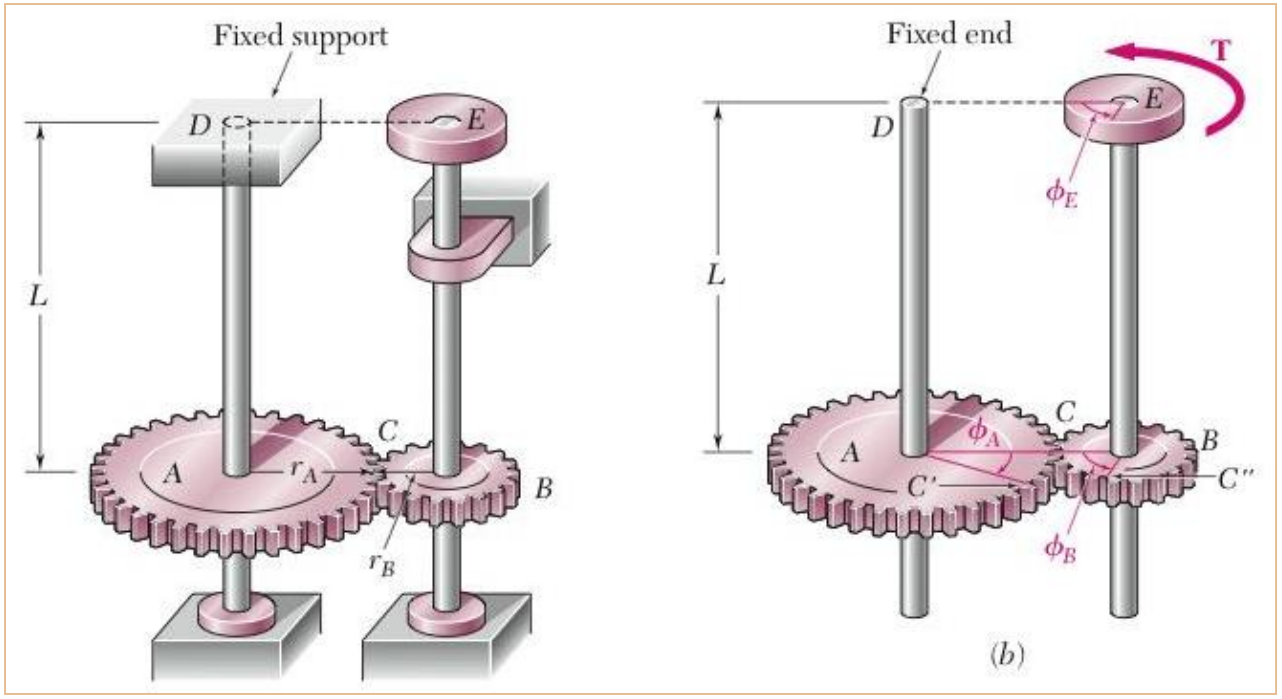




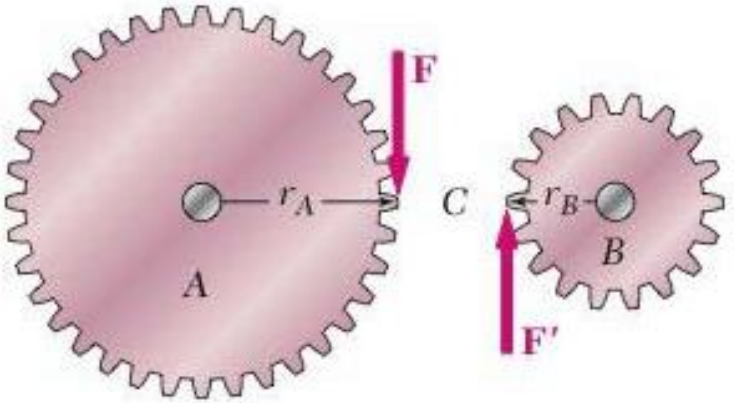


- Angle of twist is equal to the angle of rotation of the free end

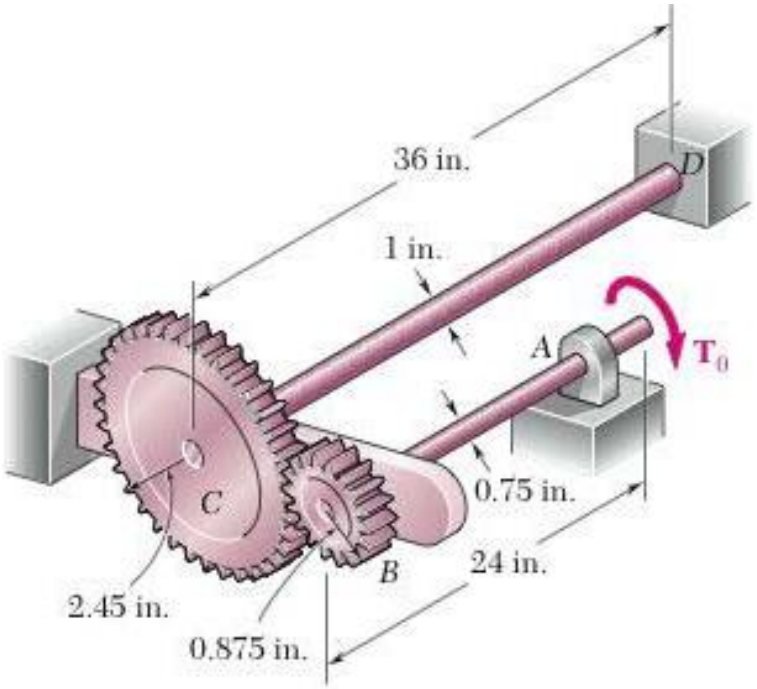
- Angle of twist is equal to the angle through which one end of the shaft rotates with respect to the other


Fixed support
D
E
L
A
C
B
r_A
r_B
Fixed end
T
D
E
ϕ_E
L
A
C
B
ϕ_A
C′
C″
ϕ_B
(b)

- Find Φ_E


F
r_A
A
C
r_B
B
F′

Problems



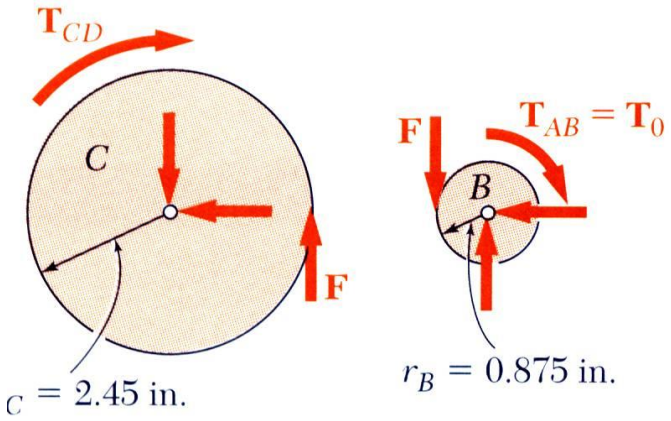
SOLUTION:

- Apply a static equilibrium analysis on the two shafts to find a relationship between T_{CD} and T_0
- Apply a kinematic analysis to relate the angular rotations of the gears
- Find the maximum allowable torque on each shaft – choose the smallest
- Find the corresponding angle of twist for each shaft and the net angular rotation of end A

Two solid steel shafts are connected by the gears shown. Knowing that for each shaft $G = 11.2 \times 10^6$ psi and that the allowable shearing stress is 8 ksi, determine (*a*) the largest torque $\mathbf{T}_0$ that may be applied to end A of shaft AB, (*b*) the corresponding angle through which end A of shaft AB rotates.

SOLUTION:

- Apply a static equilibrium analysis on the two shafts to find a relationship between T_{CD} and T_0



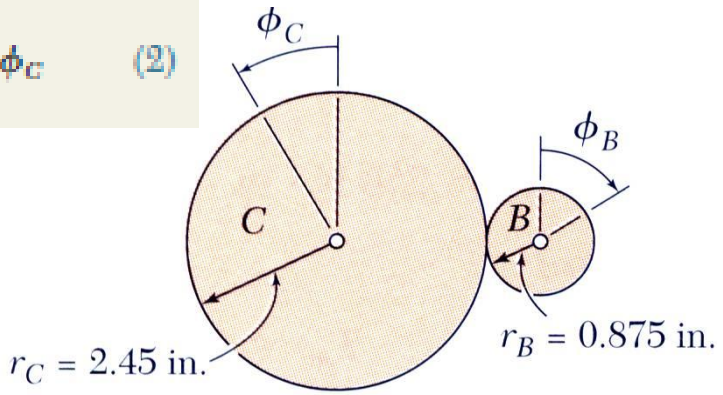
Statics. Denoting by F the magnitude of the tangential force between gear teeth, we have

$$\textit{Gear B.}\quad \Sigma M_B = 0: \qquad F(0.875\text{ in.}) - T_0 = 0$$
$$\textit{Gear C.}\quad \Sigma M_C = 0: \qquad F(2.45\text{ in.}) - T_{CD} = 0$$
$$T_{CD} = 2.8T_0 \qquad (1)$$

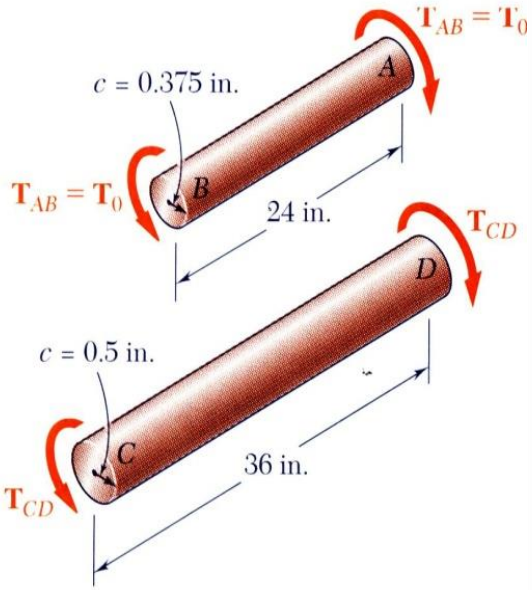
- Apply a kinematic analysis to relate the angular rotations of the gears

Kinematics. Noting that the peripheral motions of the gears are equal, we write

$$r_B\phi_B = r_C\phi_C \qquad \phi_B = \phi_C\frac{r_C}{r_B} = \phi_C\frac{2.45\text{ in.}}{0.875\text{ in.}} = 2.8\phi_C \qquad (2)$$



- Find the T_0 for the maximum allowable torque on each shaft – choose the smallest



a. Torque T_0

Shaft AB. With $T_{AB} = T_0$ and $c = 0.375$ in., together with a maximum permissible shearing stress of 8000 psi, we write

$$\tau = \frac{T_{AB}c}{J} \qquad 8000 \text{ psi} = \frac{T_0(0.375 \text{ in.})}{\frac{1}{2}\pi(0.375 \text{ in.})^4} \qquad T_0 = 663 \text{ lb} \cdot \text{in.}$$

Shaft CD. From (1) we have $T_{CD} = 2.8T_0$. With $c = 0.5$ in. and $\tau_{all} = 8000$ psi, we write

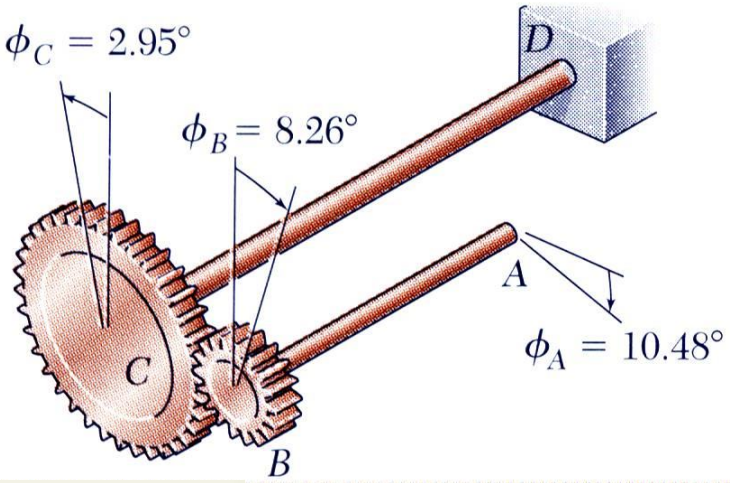
$$\tau = \frac{T_{CD}c}{J} \qquad 8000 \text{ psi} = \frac{2.8T_0(0.5 \text{ in.})}{\frac{1}{2}\pi(0.5 \text{ in.})^4} \qquad T_0 = 561 \text{ lb} \cdot \text{in.}$$

Maximum Permissible Torque. We choose the smaller value obtained for T_0

$$T_0 = 561 \text{ lb} \cdot \text{in.}$$

$$T_0 = 561\,\text{lb} \cdot \text{in}$$

- Find the corresponding angle of twist for each shaft and the net angular rotation of end *A*



b. Angle of Rotation at End *A*. We first compute the angle of twist for each shaft.

Shaft AB. For $T_{AB} = T_0 = 561 \text{ lb} \cdot \text{in.}$, we have

$$\phi_{A/B} = \frac{T_{AB}L}{JG} = \frac{(561 \text{ lb} \cdot \text{in.})(24 \text{ in.})}{\frac{1}{2}\pi(0.375 \text{ in.})^4(11.2 \times 10^6 \text{ psi})} = 0.0387 \text{ rad} = 2.22°$$

Shaft CD. $T_{CD} = 2.8T_0 = 2.8(561 \text{ lb} \cdot \text{in.})$

$$\phi_{C/D} = \frac{T_{CD}L}{JG} = \frac{2.8(561 \text{ lb} \cdot \text{in.})(36 \text{ in.})}{\frac{1}{2}\pi(0.5 \text{ in.})^4(11.2 \times 10^6 \text{ psi})} = 0.0514 \text{ rad} = 2.95°$$

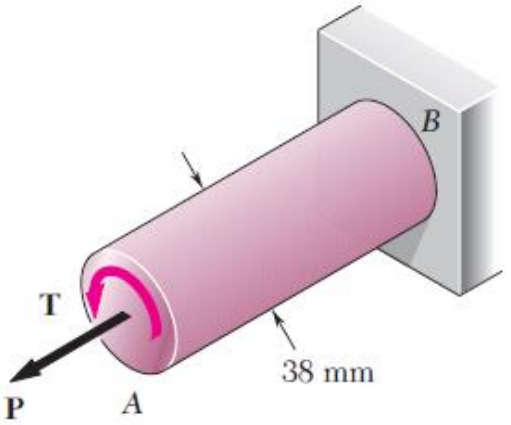
Since end *D* of shaft *CD* is fixed, we have $\phi_C = \phi_{C/D} = 2.95°$. Using (2), we find the angle of rotation of gear *B* to be

$$\phi_B = 2.8\phi_C = 2.8(2.95°) = 8.26°$$

For end *A* of shaft *AB*, we have

$$\phi_A = \phi_B + \phi_{A/B} = 8.26° + 2.22° \qquad \phi_A = 10.48°$$

The 38 mm diameter shaft AB is made of a grade of steel for which the yield strength is $Y = 250$ MPa. Using the maximum shearing stress criterion (MSS), determine the magnitude of torque T for which yield occurs when $P = 240$ kN



$$P = 240 \times 10^3 \text{ N}$$

$$A = \frac{\pi}{4} d^2 = \frac{\pi}{4}(38)^2 = 1.1341 \times 10^3 \text{ mm}^2 = 1.1341 \times 10^{-3} \text{ m}^2$$

$$\sigma_x = \frac{P}{A} = \frac{240 \times 10^3}{1.1341 \times 10^{-3}} = 211.6 \times 10^6 \text{ Pa} = 211.6 \text{ MPa}$$

$$\sigma_y = 0 \qquad \sigma_{ave} = \frac{1}{2}(\sigma_x + \sigma_y) = \frac{1}{2}\sigma_x$$

$$R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = \sqrt{\frac{1}{4}\sigma_x^2 + \tau_{xy}^2}$$

$$2\tau_{max} = 2R = \sqrt{\sigma_x^2 + 4\tau_{xy}^2} = \sigma_y$$

$$4\tau_{xy}^2 = \sigma_y^2 - \sigma_x^2 \qquad \tau_{xy} = \frac{1}{2}\sqrt{\sigma_y^2 - \sigma_x^2} = \frac{1}{2}\sqrt{250^2 - 211.6^2}$$

$$= 65.568 \times 10^6 \text{ Pa} = 66.568 \text{ MPa}$$

From torsion $\tau_{xy} = \dfrac{Tc}{J}$ $\quad T = \dfrac{J\tau_{xy}}{c}$

$$J = \frac{\pi}{2}c^4 = \frac{\pi}{2}\left(\frac{38}{2}\right)^4 = 204.71 \times 10^3 \text{ mm}^4 = 204.71 \times 10^{-9} \text{ m}^4$$

$$c = \frac{1}{2}d = 19 \times 10^{-3} \text{ m}$$

$$T = \frac{(204.71 \times 10^{-9})(65.668 \times 10^6)}{19 \times 10^{-3}} = 717 \text{ N} \cdot \text{m}$$

The rotor shaft of an helicopter drives the rotor blades that provide the lifting force to support the helicopter in the air. As a consequence, the shaft is subjected to a combination of torsion and axial loading.

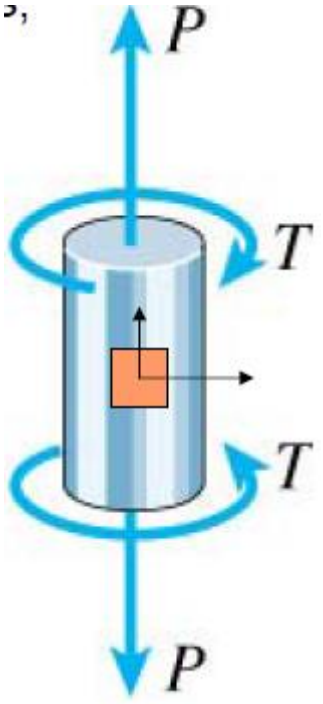
For a ***50mm*** diameter shaft transmitting a torque T **= 2.4kN.m** and a tensile force **P = 125kN**, determine the maximum tensile stress, maximum compressive stress, and maximum shear stress in the shaft.



The tensile stress

$$\sigma = \frac{P}{A} = \frac{125kN}{\pi/4(0.05m)^2} = 63.66MPa$$

$$\tau_{Torsion} = \frac{Tr}{I_P} = \frac{(2.4kN.m)\left(\frac{0.05}{2}\right)}{\frac{\pi(0.05)^4}{32}} = 97.78MPa$$



$$\sigma_{1,2} = \left(\frac{\sigma_x + \sigma_y}{2} \right) \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2} \right)^2 + (\tau_{xy})^2}$$

$$\sigma_{1,2} = \left(\frac{0 + 63.66}{2} \right) \pm \sqrt{\left(\frac{0 - 63.66}{2} \right)^2 + (-97.78)^2}$$

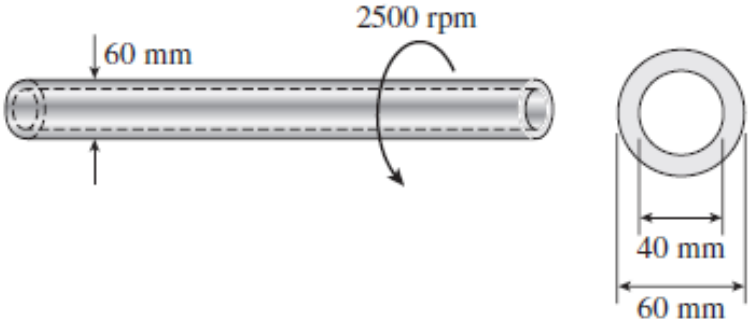
$$\sigma_1 = 135 MPa$$

$$\sigma_2 = -71 MPa$$

The drive shaft for a truck (outer diameter 60 mm and inner diameter 40 mm) is running at 2500 rpm (see figure).

(a) If the shaft transmits 150 kW, what is the maximum shear stress in the shaft?

(b) If the allowable shear stress is 30 MPa, what is the maximum power that can be transmitted?



$d_2 = 60$ mm $\quad d_1 = 40$ mm $\quad n = 2500$ rpm

$$I_P = \frac{\pi}{32}(d_2^4 - d_1^4) = 1.0210 \times 10^{-6}\ \text{m}^4$$

(a) MAXIMUM SHEAR STRESS τ_{max}

P = power (watts) $\quad P = 150$ kW $= 150{,}000$ W

T = torque (newton meters) $\quad n$ = rpm

$$P = \frac{2\pi nT}{60} \qquad T = \frac{60P}{2\pi n}$$

$$T = \frac{60(150{,}000\ \text{W})}{2\pi(2500\ \text{rpm})} = 572.96\ \text{N} \cdot \text{m}$$

$$\tau_{max} = \frac{Td_2}{2I_P} = \frac{(572.96\ \text{N} \cdot \text{m})(0.060\ \text{m})}{2(1.0210 \times 10^{-6}\ \text{m}^4)}$$

$$= 16.835\ \text{MPa}$$

$$\tau_{max} = 16.8\ \text{MPa} \quad \leftarrow$$

(b) MAXIMUM POWER P_{max}

$$\tau_{allow} = 30\ \text{MPa}$$

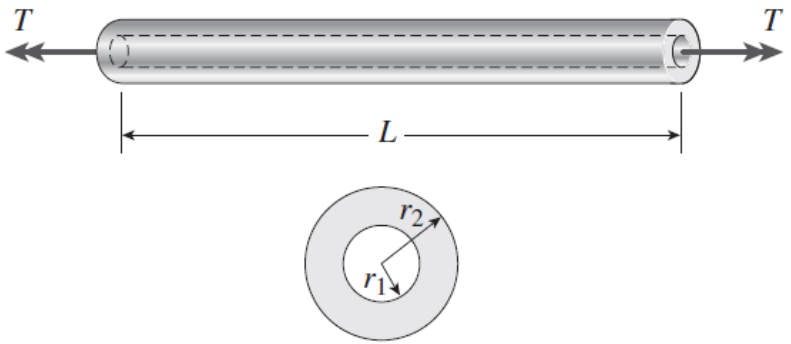
$$P_{max} = P\frac{\tau_{allow}}{\tau_{max}} = (150\ \text{kW})\left(\frac{30\ \text{MPa}}{16.835\ \text{MPa}}\right)$$

$$= 267\ \text{kW} \quad \leftarrow$$

Problem A circular aluminum tube subjected to pure torsion by torques T (see figure) has an outer radius r_2 equal to 1.5 times the inner radius r_1.

(a) If the maximum shear strain in the tube is measured as 400×10^{-6} rad, what is the shear strain γ_1 at the inner surface?

(b) If the maximum allowable rate of twist is 0.125 degrees per foot and the maximum shear strain is to be kept at 400×10^{-6} rad by adjusting the torque T, what is the minimum required outer radius $(r_2)_{min}$?



$r_2 = 1.5r_1 \quad \gamma_{max} = 400 \times (10^{-6})$ radians

$$\theta = 0.125\left(\frac{\pi}{180}\right)\left(\frac{1}{12}\right)$$

$$\theta = 1.818 \times 10^{-4} \text{ rad/m.}$$

(a) SHEAR STRAIN AT INNER SURFACE AT RADIUS r_1

$$\gamma_1 = \frac{r_1}{r_2}\gamma_{max} \qquad \gamma_1 = \frac{1}{1.5}\gamma_{max}$$

$$\gamma_1 = 267 \times 10^{-6} \text{ radians} \leftarrow$$

(b) MIN. REQUIRED OUTER RADIUS

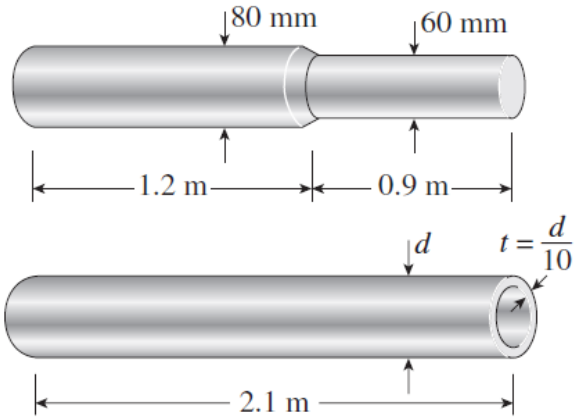
$$r_{2min} = \frac{\gamma_{max}}{\theta} \qquad r_{2min} = \frac{\gamma_{max}}{\theta}$$

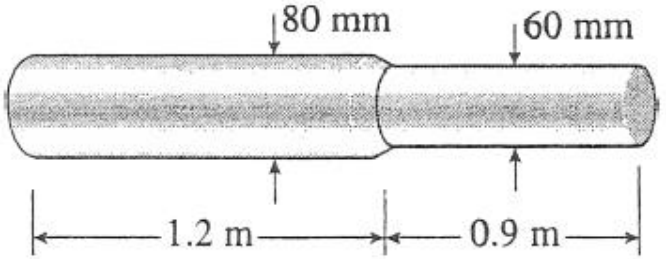
$$r_{2min} = 2.2 \text{ inches} \leftarrow$$

Problem A shaft of solid circular cross section consisting of two segments is shown in the first part of the figure. The left-hand segment has diameter 80 mm and length 1.2 m; the right-hand segment has diameter 60 mm and length 0.9 m.

Shown in the second part of the figure is a hollow shaft made of the same material and having the same length. The thickness t of the hollow shaft is $d/10$, where d is the outer diameter. Both shafts are subjected to the same torque.

If the hollow shaft is to have the same torsional stiffness as the solid shaft, what should be its outer diameter d?



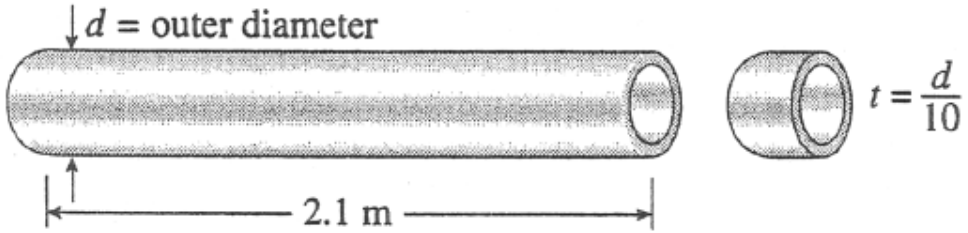


$$\phi_1 = \Sigma \frac{TL_i}{GI_{Pi}} = \frac{T(1.2\text{ m})}{G\left(\frac{\pi}{32}\right)(80\text{ mm})^4} + \frac{T(0.9\text{ m})}{G\left(\frac{\pi}{32}\right)(60\text{ mm})^4}$$

$$= \frac{32T}{\pi G}(29{,}297\text{ m}^{-3} + 69{,}444\text{ m}^{-3})$$

$$= \frac{32T}{\pi G}(98{,}741\text{ m}^{-3})$$

Hollow shaft



d_0 = inner diameter = $0.8d$

$$\phi_2 = \frac{TL}{GI_p} = \frac{T(2.1\text{ m})}{G\left(\frac{\pi}{32}\right)[d^4 - (0.8d)^4]}$$

$$= \frac{32T}{\pi G}\left(\frac{2.1\text{ m}}{0.5904\,d^4}\right) = \frac{32T}{\pi G}\left(\frac{3.5569\text{ m}}{d^4}\right)$$

TORSIONAL STIFFNESS

$$k_T = \frac{T}{\phi}$$ Torque T is the same for both shafts.

$\therefore$ For equal stiffnesses, $\phi_1 = \phi_2$

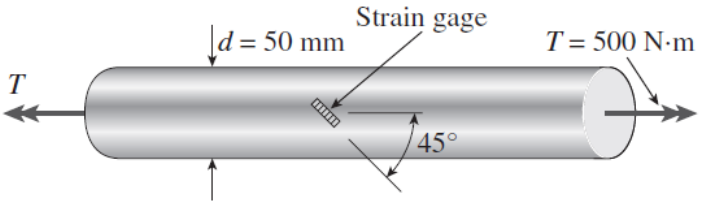
$$98{,}741\ \text{m}^{-3} = \frac{3.5569\ \text{m}}{d^4}$$

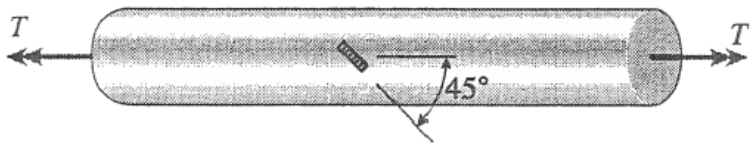
$$d^4 = \frac{3.5569}{98{,}741} = 36.023 \times 10^{-6}\ \text{m}^4$$

$$d = 0.0775\ \text{m} = 77.5\ \text{mm} \quad \leftarrow$$

A solid circular bar of diameter $d = 50$ mm (see figure) is twisted in a testing machine until the applied torque reaches the value $T = 500$ N · m. At this value of torque, a strain gage oriented at 45° to the axis of the bar gives a reading $\epsilon = 339 \times 10^{-6}$.

What is the shear modulus G of the material?





Strain gage at 45°:

$\varepsilon_{max} = 339 \times 10^{-6}$

$d = 50$ mm

$T = 500\ \text{N} \cdot \text{m}$

SHEAR STRAIN

$\gamma_{max} = 2\varepsilon_{max} = 678 \times 10^{-6}$

SHEAR STRESS

$$\tau_{max} = \frac{16T}{\pi d^3} = \frac{16(500\ \text{N} \cdot \text{m})}{\pi(0.050\ \text{m})^3} = 20.372\ \text{MPa}$$

SHEAR MODULUS

$$G = \frac{\tau_{max}}{\gamma_{max}} = \frac{20.372\ \text{MPa}}{678 \times 10^{-6}} = 30.0\ \text{GPa} \quad \leftarrow$$